Product Requirements Document

Product Requirements Document | Version 1.0 | 19 October 2026

1. Summary

MeasureMe estimates a person's standing height from one smartphone photo, entirely on the phone. It offers three measuring modes that trade convenience for accuracy. The product's defining choice is honesty: every result shows a range, and no mode claims more accuracy than the physics allows.

> **Accuracy stance.** A photo contains angles, not distances, so scale has to come from somewhere: a population prior (Mode A), a known camera height (Mode B), or a known object (Mode C). The stated targets below are hypotheses to be validated on real data, not promises.

2. Problem statement

Single-image height estimation is ill-posed. A 150 cm person close to the lens and a 190 cm person farther away can produce the same silhouette. Without something known in the scene (an object of known size, the camera's height, or the focal length plus a known ground plane), scale is ambiguous. MeasureMe attacks that ambiguity three different ways.

3. Target audience

- **High school and college students (14-22).** Curiosity and quick checks. Low patience, so the flow must work on the first try.
- **Athletes.** Repeated measurements over months. They value consistency between sessions as much as absolute accuracy.
- **Parents tracking growth.** Children measured every few weeks. The app is not a medical device and says so. Photos of minors raise privacy obligations. On-device processing helps, but check Google Play's policies for apps serving children and your local privacy law. Insufficient data to verify the specific requirements.

4. Core features: the three modes

Mode	User provides	Scale comes from	Expected error (hypothesis, unvalidated)
A. Proportions	One photo. Optional sex and age.	Population prior on head size and body proportions	Several cm, wide range. Cannot beat the sex/age baseline by much.
B. Horizon	One photo, phone tilt, camera height	Camera height plus ground-plane geometry	Low single-digit cm if camera height is accurate
C. Reference	One photo with an A4 sheet flat on the floor beside the feet	A4 size (210 x 297 mm) gives camera height	Best of the three

> **Design decision.** Mode C measures the *floor*, not the person. The A4 sheet's corners give a homography, which with the phone's focal length gives the camera's height above the floor. That height feeds the Mode B formula. Modes B and C share one geometry engine and differ only in where camera height comes from. A sheet held upright against the body is a possible later variant but is more sensitive to depth error.

5. User flow

Home, then Capture, then Processing, then Result.

Step	What the user sees	System behavior
Home	Three mode cards, each stating what it needs and how accurate it typically is. Continue button. Privacy line.	Stores last-used mode. Optional sex and age inputs (on-device only).
Capture	Viewfinder with corner brackets and body guide. Checklist of three tips. Take photo, Choose photo, Use a sample. Measure my height (disabled until a photo exists).	Live quality hints: feet in frame, phone level, subject centered, enough light. Shutter unlocks only when the quality gate passes.
Processing	Progress bar and three steps: find person, locate head and feet, work out the scale.	Under 3 seconds on a mid-range phone. Steps reflect real pipeline stages. No fake progress.
Result	Height with a count-up, imperial value, range, photo-quality bar, plain-language caveat, stats, Measure again.	Saves to local history (P1). Failure paths return to Capture with one fixable reason.

6. Functional requirements

IDs are stable so tests and tasks can reference them.

ID	Priority	Requirement
FR-1	P0	All inference runs on-device. No account. No network call carries image data.
FR-2	P0	A quality gate runs before processing and returns one specific, fixable reason on failure.
FR-3	P0	Every result shows a point estimate and a calibrated range (80% target coverage).
FR-4	P0	Heights display in cm and ft/in.
FR-5	P0	A visible note states the app is not a medical measurement.
FR-6	P0	Mode A (Proportions) and Mode C (Reference) work end to end.
FR-7	P1	Mode B (Horizon) with user-entered or Mode-C-derived camera height.
FR-8	P1	History and growth chart, stored locally (Room).
FR-9	P1	Manual landmark adjustment ("Adjust points").
FR-10	P1	Optional sex and age input that improves the Mode A prior.
FR-11	P1	CSV export of history.
NG-1	Non-goal	Medical-grade accuracy, infants or anyone not standing, multi-person photos.

7. Success metrics

7.1 Accuracy

Metric	Definition	Target
MAE	Mean absolute error vs. stadiometer	A: 5 cm or less on adults passing the quality gate. B: 5 cm or less. C: 3 cm or less.
Skill score	1 - (MAE_model / MAE_baseline), baseline = mean height for the person's sex and age	A: 0.25 or higher
P90 error	90th percentile of absolute error	Report. No target until data exists.
Interval coverage	Share of true heights inside the stated 80% range	75-85%
Bias	Mean signed error by height bucket	Within 1.5 cm

> **Why a skill score.** For normally distributed heights, guessing the mean has MAE of about 0.8 x SD. With an adult SD of roughly 6.5-7 cm (approximate, from general anthropometric knowledge; verify before quoting), that baseline is already about 5-5.6 cm, and worse for mixed-age teens. A 5 cm MAE for Mode A with skill near zero is not a useful product. Report MAE and skill together.

7.2 Validation protocol

- At least 100 subjects, at least 3 photos each, at least 3 phone models.
- Ground truth from a stadiometer or wall-mounted tape, barefoot, measured by another person. Real stature varies by roughly 1-2 cm over a day, so photograph and measure within the same hour.
- Split train and test by subject, never by photo. Photos of one person on both sides inflate results.
- Report errors per mode, per phone, and per quality bucket.

7.3 Product

- First-try pass rate of the quality gate: 70% or higher.
- Flow completion rate, Home to Result.
- Retention at D1, D7 and D30. Hypothesis: growth tracking drives D30. Test it before building around it.
- Repeat-measurement rate per user.

8. Risks

Risk	Mitigation
Over-promising accuracy	Ranges, honest copy, skill-score metric, in-app caveat.
Privacy with minors	On-device only, no cloud, no analytics on image content. Review Play policy for child-directed apps.
Phone intrinsics vary (focal length, crop, digital zoom)	Read Camera2 characteristics, validate per phone model.
Ground truth noise	Same-hour measurement, barefoot, a second measurer.

9. Open questions

1. How does the user enter camera height in Mode B without it feeling like homework? Options: type a number, pick a preset (tripod, shelf, chair), or derive it from a Mode C sheet.
2. Is sex and age input acceptable for the teen and parent audience, given it improves accuracy but adds friction?
3. Does the app offer a manual crown and heel adjustment in v1, or only in a later release?